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Raw Materials (InP, Ge, LiNbO3)

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Substrate (AXTI, IQE, Sumitomo)
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Epitaxy/Wafer (IQE, Coherent)
↓
Chip Fab (TSEM, GFS, TSMC)
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Components (LITE, COHR, MACOM)
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Packaging/CPO (TSMC COUPE, FN)
↓
Systems (NVDA, AVGO, MRVL)
↓
Hyperscalers (Google, Microsoft, AWS, Meta)

tìm tất cả thông tin liên quan đến những mã này trong lĩnh vực photonic AI

Đây là tổng hợp chi tiết về tất cả các mã/công ty trong chuỗi cung ứng **Photonic AI** (quang học tích hợp cho AI), được sắp xếp theo từng tầng trong sơ đồ bạn cung cấp. Photonic AI đang trở thành công nghệ then chốt để giải quyết bài toán băng thông và tiêu thụ năng lượng trong các cụm AI lớn.

Tổng quan: Tại sao Photonic AI quan trọng?

AI cần truyền tải lượng dữ liệu khổng lồ giữa các GPU/TPU. Công nghệ quang học (sử dụng ánh sáng thay vì đồng) giúp:

- **Tốc độ cao hơn**: Lên đến 224 Gbps/lane [1]
- **Tiết kiệm năng lượng**: Giảm từ 15-20 pJ/bit xuống còn 5 pJ/bit với CPO [2]
- **Băng thông lớn hơn**: 10x increase trong bandwidth density [3]

1️⃣ Raw Materials (Vật liệu thô)

Vật liệu	Vai trò trong Photonic AI
InP (Indium Phosphide)	Vật liệu chủ động cho laser, bộ khuếch đại, phát hiện ánh sáng; nền tảng cho optical links tốc độ cao up to 200 Gbps/lane [4] [5]
Ge (Germanium)	Dùng trong photodetectors cho silicon photonics
$LiNbO_3$ (Lithium Niobate)	Vượt trội trong điều chế chủ động (electro-optic modulator); photonic processor dựa trên $LiNbO_3$ mang lại 30x efficiency improvement cho AI so với digital CMOS [6] [7]

2️⃣ Substrate (Đế bán dẫn)

Công ty	Mã	Vai trò
AXTI	$AXTI	Nhà cung cấp hàng đầu InP substrate; trực tiếp phục vụ AI, data center connectivity (silicon photonics), 5G [8] [9]
IQE	$IQEPF	Cung cấp substrate và epitaxy; leader epitaxy phương Tây [8] [10]
Sumitomo	(Tập đoàn Nhật)	Cung cấp InP substrate; phát triển VCBEL Flip-Chip Integration on Glass cho CPO [11] [9]

3️⃣ Epitaxy/Wafer (Lớp phim mỏng)

Công ty	Mã	Vai trò
IQE	$IQEPF	Leader epitaxy phương Tây; cung cấp epitaxial wafers cho quantum dot laser và SOA cho AI optical interconnects [12] [10]
Coherent	$COHR	Cung cấp epitaxy + silicon photonics transceivers cho AI data centers [10] [13]

Lưu ý: IQE và Quintessent hợp tác xây dựng chuỗi cung ứng epitaxial wafer cho quantum dot laser phục vụ AI optical interconnects.[12]

4️⃣ Chip Fab (Sản xuất chip)

Công ty	Mã	Vai trò trong Photonic AI
TSEM (Tower Semiconductor)	$TSEM	Foundry silicon photonics; hợp tác với NVIDIA phát triển optical modules 1.6T cho AI data center [14]; hợp tác Scintil Photonics cho DWDM laser sources AI [15] [16]
GFS (GlobalFoundries)	$GFS	Mở rộng vào silicon photonics cho optical interconnects AI data centers; hợp tác Flexcompute [17]
TSMC	$TSM	Phát triển COUPE (Compact Universal Photonic Engine) - nền tảng silicon photonics + CPO cho AI [18] [19]

5️⃣ Components (Linh kiện quang học)

Công ty	Mã	Vai trò
LITE (Lumentum)	$LITE	Leader optical/photonic; InP photonic solutions cho 400G/800G/1.6Tbps transceivers; hợp tác NVIDIA cho CPO tương lai [20] [21] [5]; phát triển External Laser Source modules cho CPO [22]
COHR (Coherent)	$COHR	Silicon photonics transceiver 2x400G-FR4 Lite tối ưu cho AI data centers; 1.6T/3.2T/12.8T architecture cho AI connectivity [13] [23] [24]
MACOM	$MACOM	Chipset cho 1.6 Tbps optical modules; linear driver cho 224-Gbps silicon photonics transmitter thế hệ AI tiếp theo [25] [26] [1]

6️⃣ Packaging/CPO (Đóng gói quang học tích hợp)

Công ty/Công nghệ	Vai trò
TSMC COUPE	Nền tảng CPO gồm EIC (electronic IC) + PIC (photonic IC) stacked bằng SoIC-X; roadmap 3 giai đoạn: 1.6T → 6.4T → 12.8T [18] [19] [27]
FN (Fabrinet)	System integration cho NVIDIA CPO supply chain; đóng gói/test CPO [8] [28]
FIT (Foxconn Interconnect)	Hợp tác MediaTek phát triển 51.2T CPO solution; validated 102.4T CPO External Laser Platform cho AI [29] [30]

7️⃣ Systems (Hệ thống AI)

Công ty	Mã	Vai trò trong Photonic AI
NVDA (NVIDIA)	$NVDA	leaders CPO cho AI clusters; sẽ triển khai light-based GPU interconnects từ 2026; hợp tác TSMC COUPE; Quantum-X & Spectrum-X photonics [31] [32] [33] [34]
AVGO (Broadcom)	$AVGO	Tham gia define optical scale-up interconnect cho AI clusters cùng AMD, NVIDIA và hyperscalers [35]
MRVL (Marvell)	$MRVL	Hợp tác NVIDIA NVLink Fusion (NVIDIA invest $2B); leadership optical DSP, silicon photonics, custom silicon cho AI [36] [37] [38]

8️⃣ Hyperscalers (Khách hàng cuối)

Công ty	Vai trò
Google	Leader deploy AI clusters; driver CPO cho ai training clusters [39]
Microsoft	Tham gia define optical scale-up interconnect cho AI [39] [35]
AWS (Amazon)	Hyperscale data center driver cho photonics [39]
Meta	Tham gia define optical scale-up interconnect cho AI [39] [35]

Hyperscalers là động lực chính: AI training clusters của họ có nhu cầu "insatiable" về bandwidth và power efficiency, thúc đẩy toàn bộ supply chain photonics.[39]

Mối quan hệ chiến lược quan trọng

Đối tác	Thỏa thuận
NVIDIA - Lumentum - Coherent	Hợp tác chiến lược CPO trị giá **$40 billion** tương lai [21]
NVIDIA - Marvell	NVIDIA invest **$2B** qua NVLink Fusion; hợp tác silicon photonics [37] [38]
NVIDIA - TSMC	CPO dựa trên COUPE platform của TSMC [32] [33]
Tower - NVIDIA	Strategic collaboration silicon photonics cho 1.6T optical modules [14]

Xu hướng 2025-2026

- **2025-2026:** Cửa sổ quan trọng cho pilot deployments CPO [39]
- **2026:** NVIDIA triển khai CPO cho AI clusters; 1.6T optical modules mass production [19] [27]
- **Tương lai: 100% AI data center interconnects sẽ là optical trong 5 năm** [40]
- **Thị trường:** AI Networking market đạt **$200B** vào 2030 [15]

Bạn có muốn tìm hiểu sâu hơn về một công ty/kỹ thuật cụ thể nào (ví dụ: chi tiết về COUPE technology hoặc so sánh LITE vs COHR)?

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